Gastric Cancer

Epidemiology

- 5th leading cause of cancer death globally
- ~968,350 new cases and ~659,853 deaths worldwide (2022)
- ~30,300 new cases and ~10,780 deaths in the US (2025)

(**patel342?**)

Global Epidemiology

- **Incidence hotspots:** East Asia, Eastern Europe, South America
- Incidence 2× higher in males than females
- Median age at diagnosis: **68 years**
- Globally, 85% of cases in stomach body or antrum; 15% in the cardia
- In North America/Western Europe, cardia/GEJ tumors are 30%–40% of cases
- Overall incidence declining over the past century
  - improved food preservation
  - declining H. pylori prevalence

(**patel342?**)

Risk Factors - Infectious

- Helicobacter pylori:
  - WHO Class I carcinogen; attributable risk of 75–90% for noncardia gastric cancer.

- Only ~3% of infected individuals develop cancer.
- Key virulence factors: CagA (oncoprotein) and VacA (vacuolating cytotoxin).

Epstein-Barr virus (EBV): associated with ~7.5% of gastric cancers. **(patel342?)**

Risk Factors - Environmental/Lifestyle

- High salt intake, smoked/preserved foods (OR ~2.1) [5]
- Smoking (OR ~1.25), heavy alcohol use (OR ~1.37) [1]
- Obesity (OR ~1.7) — particularly for cardia/GEJ cancers [5]

(patel342?)

Risk Factors - Medical Conditions

- Pernicious anemia / autoimmune atrophic gastritis [3]
- Prior partial gastrectomy (gastric stump cancer)
- Gastric adenomatous polyps [6]
- Ménétrier disease (giant hypertrophic gastritis)

Risk Factors - Genetic/Hereditary

- CDH1 mutations → Hereditary Diffuse Gastric Cancer (HDGC)
  - autosomal dominant; 37–42% lifetime risk in males, 25–33% in females.
  - CDH1 encodes E-cadherin, a cell adhesion protein.
  - Also associated with lobular breast cancer (37–55% lifetime risk in females).[1]

[1] **Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, Esophageal, and Gastric.**

- Lynch syndrome (mismatch repair gene mutations) — gastric cancer in up to 10% of families
- FAP (APC mutations) — increased risk of gastric adenomas and cancer
- BRCA1/BRCA2 germline variants — emerging evidence of increased gastric cancer risk

Protective Factors

- H. pylori eradication (46% reduction in gastric cancer incidence) [2]
- Diets high in fruits and vegetables
- Aspirin/NSAID use (some evidence)

Histologic Types

Feataures	Intestinal Type (~50%)	Diffuse Type (~30%)
Histology	Glandular/papillary well-differentiated	Poorly cohesive, dedifferentiated, **signet ring cells**
Precursor	Follows Correa cascade	Does NOT require chronic inflammation
H Pylori Association	Strong	Present
Location	Antrum, body	Any location (or entire stomach)
Gross appearance	Polypoid, fungating	Linitis plastica ("leather bottle")
Spread	Hematogenous (Liver)	Peritoneal (Krukenburg tumors, carcinomatosis)
Key mutation	P53	**CDH1** (E-cadherin loss)

[2](**chey1730?**)

Mixed tumors in 20%

Signet ring cell histology portends resistance to systemic therapy and poor prognosis

(**patel342?**)

Adenocarcinoma GE Junction - Siewert Classification

Type I: Distal esophagus

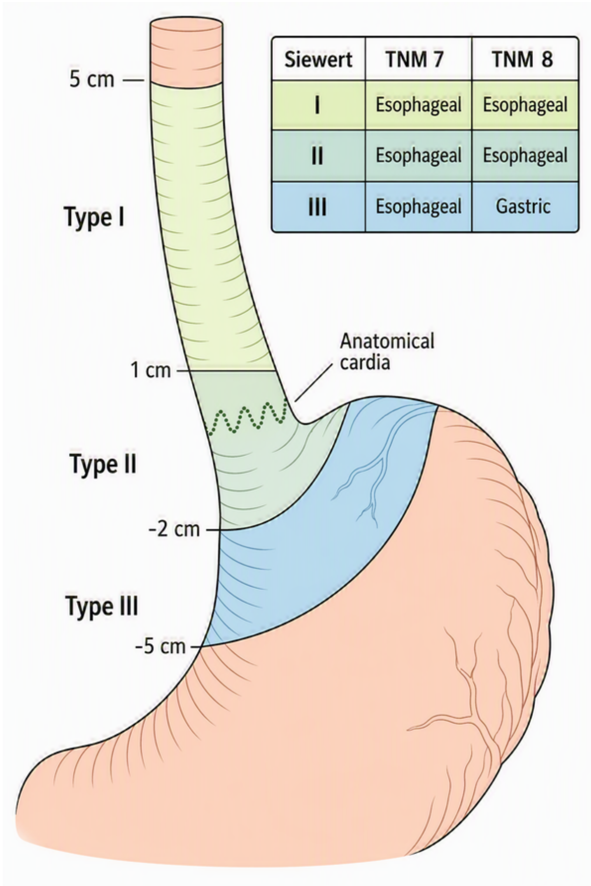
- (1–5 cm above GEJ)

Type II: True cardia

- (1 cm above to 2 cm below GEJ)

Type III: Subcardial

- (2–5 cm below GEJ)



Molecular Genetics

Subtype Frequency Key Features Clinical Relevance Chromosomal Instability (CIN) ~50% TP53 mutations, aneuploidy, RTK amplifications (HER2, EGFR, FGFR2) Most common; responds to cisplatin-based chemo; HER2-targeted therapy Microsatellite Instability (MSI) ~20% High mutation burden, MLH1 hypermethylation, dense lymphocyte infiltration Better prognosis; responds to immune checkpoint inhibitors EBV-positive ~9% PIK3CA mutations, PD-L1/PD-L2 amplification, extreme DNA hypermethylation High immunogenicity; responds to immunotherapy Genomically Stable (GS) ~20% CDH1/RHOA mutations, CLDN18-ARHGAP fusions, diffuse histology Worst prognosis; target for CLDN18.2-directed therapy

Gastric Cancer Treatment Categories

Category	AJCC	Stage	Treatment
Superficial Tumors	I	T1a	Endoscopic Therapy
Localized Tumors	IIB	T1b T2	Surgery
Locally-advanced	III IIA	T3 or N^+	Chemo $\rightarrow$ Surgery$\rightarrow$ Chemo
Metastatic	IVB	M1	Chemotherapy $\pm$Radiation

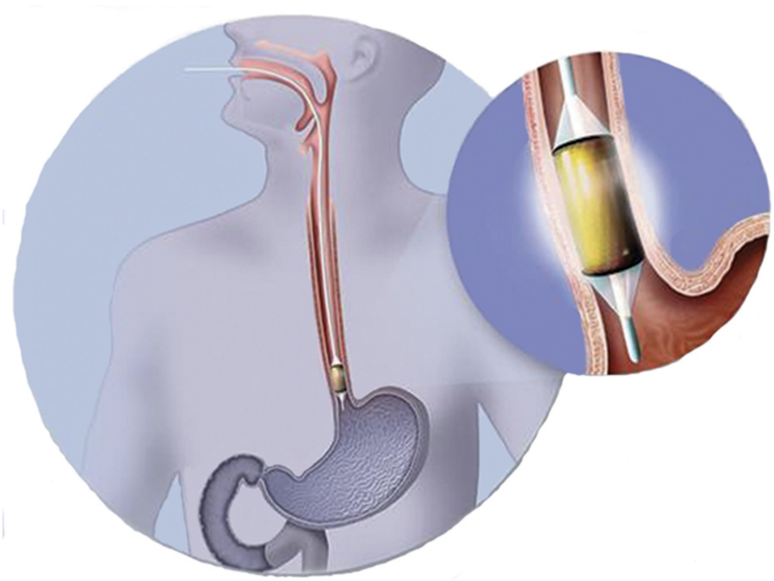
Superficial Gastric Cancer

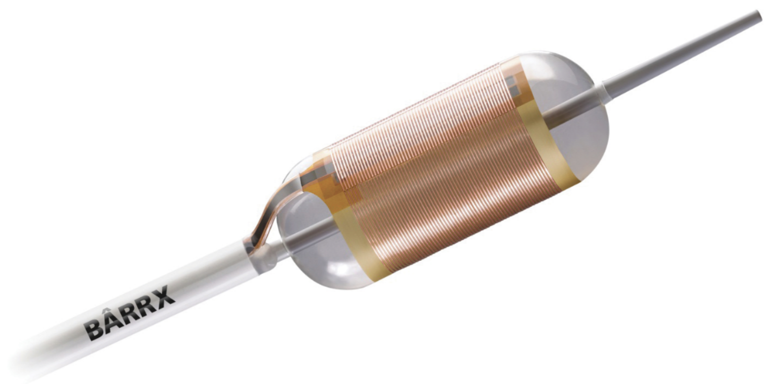
127 patients with dysplasia randomized:

- Radio-frequency ablation
- Sham ablation

Low-grade dysplasia in 64

High-grade dysplasia in 63

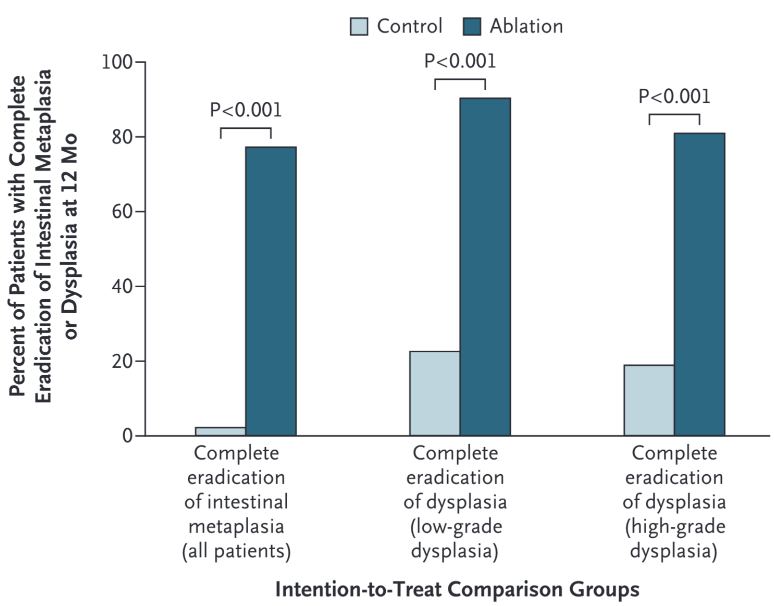




(Shaheen et al. 2009)

Radiofrequency Ablation for Dysplasia

Radiofrequency Ablation results in eradication of Barrett's in 75% at 1 year



No role for primary surgery in high-grade dysplasia

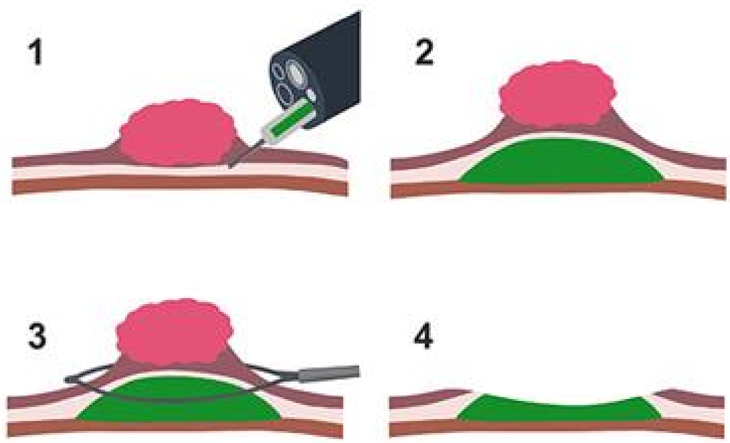
(Shaheen et al. 2009)

Superficial Tumors (T1)

Workup of nodular Barretts:

- Endoscopic Ultrasound
- Endoscopic Mucosal Resection
  - Diagnostic (T staging)
  - May be therapeutic for T1a tumors

Endoscopic Musocal Resection for T1 Tumors



Endoscopic Submucosal Dissection for T1a/T1b

Endoscopic resection uses needle-knife to dissect *below* submucosa

Maybe suitable for T1b lesions *if* lesion is completely resected

Risk of perforation

Localized Tumors (T2 N0)

SLIDE 2: Learning Objectives

By the end of this presentation, students should be able to:

1. Describe the epidemiology and key risk factors for gastric cancer
2. Explain the Correa cascade and pathogenesis of gastric cancer
3. Differentiate histologic (Laurén) and molecular (TCGA) classification systems
4. Recognize the clinical presentation and diagnostic workup
5. Understand the TNM staging system and its prognostic implications
6. Outline stage-appropriate treatment strategies including surgery, chemotherapy, immunotherapy, and targeted therapy
7. Identify key biomarkers that guide treatment decisions

Intestinal-type gastric carcinogenesis: Corea Cascase
Normal mucosa → Chronic gastritis (H. pylori) → Atrophic gastritis → Intestinal metaplasia → Dysplasia → Adenocarcinoma
Each step involves accumulating genetic and epigenetic changes (chromosomal instability, DNA methylation). [3][11]
Gastric intestinal metaplasia (GIM) is a key precancerous lesion; 10-year cumulative risk of progression to cancer is ~1.6% overall, but 2–4.5× higher in high-risk GIM. [11]
H. pylori eradication can reduce cancer risk but has limited effect on reversing established intestinal metaplasia. [12]
SLIDE 8: Molecular Classification — TCGA Subtypes

Subtype	Frequency	Key Features	Clinical Relevance
Chromosomal Instability (CIN)	~50%	TP53 mutations, aneuploidy, RTK amplifications (ERBB2, EGFR, FGFR2)	80% of GEJ/cardia tumors; HER2-targeted therapy
Microsatellite Instability (MSI)	~20%	High mutation burden, dMMR, dense lymphocyte infiltration	Best response to immune checkpoint inhibitors
EBV-positive	~9%	PIK3CA mutations, PD-L1/PD-L2 overexpression, CDKN2A methylation	High immunogenicity; responds to immunotherapy
Genomically Stable (GS)	~20%	CDH1/RHOA mutations, diffuse histology, EMT features	Worst prognosis; limited targeted options

- MSI and EBV-positive subtypes are associated with better responses to immune checkpoint inhibitors - CIN subtype enriched for targetable amplifications (HER2, FGFR2)

SLIDE 9: Clinical Presentation

Most patients are symptomatic at diagnosis (>90% in the US)

Common symptoms:

- Weight loss (62%)
- Abdominal/epigastric pain (52%)
- Nausea (34%)
- Anorexia/early satiety (32%)
- Dysphagia (26%) — especially proximal/GEJ tumors
- Iron deficiency anemia (40% in one series)

Late/advanced findings:

- Virchow node (left supraclavicular lymphadenopathy)
- Sister Mary Joseph nodule (periumbilical nodule)
- Krukenberg tumor (ovarian metastasis)
- Blumer shelf (peritoneal drop metastasis on rectal exam)
- Ascites (peritoneal carcinomatosis)

Key point: Early gastric cancer is usually asymptomatic — this is why screening is effective in high-incidence countries

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SLIDE 10: Screening

High-incidence countries (Japan, South Korea):

- Population-based endoscopic screening every 2 years
  - South Korea: age 40 years
  - Japan: age 50 years
- South Korea: screening participation increased from 39.4% (2005) to 77.5 (2023)
- Localized-stage detection increased from 51.7% to 69.8%
- 5-year relative survival improved from 58% to 78.4%
- Japanese cohort study: 61% reduction in gastric cancer mortality with endoscopic screening (HR 0.39)

United States:

- No routine screening for general population (low incidence)
- AGA (2025): screening endoscopy recommended for high-risk populations (foreign-born individuals from moderate-to high-incidence regions)
- Endoscopy recommended for new-onset dyspepsia at age 50–55 years, or any age with alarm symptoms

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SLIDE 11: Diagnostic Workup

Diagnosis:

- Upper endoscopy (EGD) with biopsy — gold standard
- 6–8 biopsies recommended for adequate histologic and molecular assessment
- Endoscopic mucosal resection (EMR) or endoscopic submucosal dissection (ESD) for small lesions 2 cm — both diagnostic and potentially therapeutic

Staging workup:

- CT chest/abdomen/pelvis with oral and IV contrast
- Endoscopic ultrasound (EUS) — best for T and N staging of early-stage disease
- PET/CT — for locally advanced disease to detect occult metastases
- Diagnostic laparoscopy with peritoneal cytology — recommended for T1b to exclude peritoneal disease

Biomarker testing (all newly diagnosed patients):

- MSI/MMR status (universal)
- PD-L1 expression (universal)
- HER2 (ERBB2) — if advanced/metastatic disease
- CLDN18.2 — if advanced/metastatic disease
- Consider NGS

SLIDE 12: TNM Staging (AJCC 8th Edition) — Simplified

T (Tumor depth):

- Tis: Carcinoma in situ
- T1a: Lamina propria/muscularis mucosae
- T1b: Submucosa
- T2: Muscularis propria
- T3: Subserosa
- T4a: Serosa (visceral peritoneum)
- T4b: Adjacent structures

N (Regional lymph nodes):

- N0: No regional LN metastasis
- N1: 1–2 regional LNs
- N2: 3–6 regional LNs
- N3a: 7–15 regional LNs
- N3b: 16 regional LNs

M (Distant metastasis):

- M0: No distant metastasis

- M1: Distant metastasis (includes positive peritoneal cytology)

Clinical stage groups:

- Stage I: T1–T2, N0, M0
- Stage IIA: T1–T2, N+, M0
- Stage IIB: T3–T4a, N0, M0
- Stage III: T3–T4a, N+, M0
- Stage IVA: T4b, any N, M0
- Stage IVB: Any T, any N, M1

SLIDE 13: Prognosis by Stage

StageDescription5-Year SurvivalTreatment IntentStage I (Localized)Confined to stomach wall~75%CurativeStage II–III (Locally advanced)Deeper invasion ± regional LNs~35%–45% (with perioperative therapy)CurativeStage IV (Metastatic)Distant metastasisPalliative / survival prolongation

- Most common metastatic sites: liver (26%–48%), peritoneum (32%–46%), distant lymph nodes (11%–20%)
- Median survival for metastatic disease: ~1–2 years with modern regimens (14–20 months)
- At presentation in the US: ~13% localized, 15%–25% locally advanced, 35%–65% metastatic

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SLIDE 14: Treatment Overview by Stage

StagePrimary Treatment**Early (Tis, T1a)**Endoscopic resection (EMR/ESD) or surgery**Localized (deep T1b, T2 N0)**Surgery (gastrectomy) ± adjuvant therapy**Locally advanced (T2 and/or N+)**Perioperative chemotherapy + surgery**Unresectable/Metastatic**Systemic therapy (chemo ± immunotherapy ± targeted therapy)

SLIDE 15: Surgery — Principles

Types of resection:

- Subtotal (distal) gastrectomy — for distal tumors
- Total gastrectomy — for proximal or diffuse tumors
- Endoscopic resection (EMR/ESD) — for select T1a lesions

Lymphadenectomy:

- **D2 lymphadenectomy** is the standard of care (perigastric + celiac artery branch nodes)
- Minimum 16 lymph nodes should be examined for adequate staging
- D1 dissection (perigastric only) is associated with higher recurrence rates

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SLIDE 16: Perioperative Chemotherapy — Locally Advanced Disease

Standard of care: Perioperative FLOT (Category 1)

- **F**luorouracil + **L**eucovorin + **O**xaliplatin + Docetaxel (**T**axotere)
- 4 cycles preoperative → surgery → 4 cycles postoperative
- FLOT4 trial: median OS 50 months vs. 35 months with ECF/ECX (HR 0.77)

FLOT + Durvalumab (anti-PD-L1):

- For PD-L1 CPS 1: Category 1 recommendation
- MATTERHORN trial: 2-year EFS 67.4% vs. 58.5% (HR 0.71)
- Note: Diffuse-type tumors did not show OS advantage with durvalumab addition

MSI-H/dMMR tumors:

- Consider neoadjuvant immunotherapy (pembrolizumab, dostarlimab, nivolumab + ipilimumab)

- High pathologic complete response rates reported
- Gastrectomy remains standard outside of organ-preservation trials

Postoperative options (if no preoperative therapy given):

- After D2 dissection: adjuvant chemotherapy (capecitabine + oxaliplatin)
- After AgentTargetSettingKey TrialTrastuzumabHER2 (ERBB2)1st-line metastaticToGAPembrolizumabPD-11st-line metastatic; perioperativeKEYNOTE-859, KEYNOTE-585NivolumabPD-11st-line metastatic; adjuvantCheckMate 649TislelizumabPD-11st-line metastaticRATIONALE-305DurvalumabPD-L1Peri-operativeMATTERHORNZolbetuximabCLDN18.21st-line metastaticSPOTLIGHT, GLOWRamucirumab-VEGFR22nd-line metastaticREGARD, RAINBOWT-DXdHER2 (ADC)2nd-line+ metastaticDESTINY-Gastric01

SLIDE 21: Hereditary Gastric Cancer Syndromes

Hereditary Diffuse Gastric Cancer (HDGC):

- CDH1 germline mutation (E-cadherin)
- Autosomal dominant
- Lifetime gastric cancer risk: 56%–70% (men), 33%–83% (women)
- Also associated with lobular breast cancer
- Management: genetic counseling, prophylactic total gastrectomy recommended for carriers
- Screening: annual EGD with random biopsies if surgery deferred

Lynch Syndrome:

- Mismatch repair gene mutations (MLH1, MSH2, MSH6, PMS2)

- Gastric cancer risk: up to 10% of families
- Screening: EGD every 2–3 years starting at age 40

Other syndromes:

- Familial adenomatous polyposis (FAP)
- Peutz-Jeghers syndrome
- Li-Fraumeni syndrome
- BRCA1/BRCA2 carriers (emerging evidence of increased risk)

SLIDE 22: Supportive Care and Nutritional Considerations

- **Nutritional assessment** is essential at diagnosis and throughout treatment
- Post-gastrectomy complications:
  - Dumping syndrome
  - Vitamin B12 deficiency (requires lifelong supplementation after total gastrectomy)
  - Iron deficiency anemia
  - Calcium/vitamin D malabsorption
  - Weight loss and malnutrition
- Early palliative care integration associated with **3 months of survival benefit**
- Psychosocial support and distress screening recommended for all patients

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SLIDE 23: H. pylori — Prevention Opportunity

- H. pylori eradication reduces gastric cancer incidence
  - Chinese RCT (180,284 individuals): HR 0.86 for gastric cancer incidence
  - Confirmed eradication: HR 0.81

- Indications for H. pylori testing and eradication:
  - Personal history of gastric neoplasia
  - Family history of gastric cancer
  - All patients with early gastric cancer
- Population-based screen-and-treat programs being evaluated in high-incidence regions
- In low-incidence areas, routine H. pylori eradication for cancer prevention is not currently recommended for individuals without other risk factors

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SLIDE 24: Key Takeaways

1. **H. pylori** is the most important modifiable risk factor (~90% of noncardia gastric cancers)
2. Most patients in the US present with **advanced disease** — early detection improves survival dramatically
3. **Molecular profiling** (HER2, PD-L1, MSI, CLDN18.2) is essential for treatment selection
4. **Perioperative FLOT** is the standard chemotherapy for locally advanced resectable disease; durvalumab may be added for PD-L1+ tumors
5. **Biomarker-driven therapy** has transformed metastatic gastric cancer management
6. **MSI-H/dMMR tumors** have the best response to immunotherapy and may not require chemotherapy
7. **Hereditary syndromes** (especially CDH1/HDGC) require genetic counseling and may warrant prophylactic gastrectomy
8. **Supportive care** and nutritional management are integral to treatment

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SLIDE 25: Review Questions

Q1: A 65-year-old man presents with weight loss and epigastric pain. EGD reveals a gastric mass. Biopsy shows adenocarcinoma. What biomarkers should be tested?

→ MSI/MMR, PD-L1, HER2, CLDN18.2

Q2: What is the most important risk factor for noncardia gastric cancer worldwide?

→ Helicobacter pylori infection

Q3: A patient with locally advanced gastric cancer (cT3N1M0) is medically fit for surgery. What is the recommended treatment approach?

→ Perioperative FLOT chemotherapy (± durvalumab if PD-L1 CPS 1) with curative-intent gastrectomy

Q4: Which TCGA molecular subtype is most responsive to immune checkpoint inhibitors?

→ MSI and EBV-positive subtypes

Q5: A 35-year-old woman is found to carry a CDH1 germline mutation. What is the recommended management?

→ Genetic counseling and consideration of prophylactic total gastrectomy

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Slide 25 — Review Questions

Q1: A 65-year-old man presents with weight loss and epigastric pain. EGD reveals a gastric body mass. Biopsy shows adenocarcinoma. CT shows no metastases. EUS shows T3N1 disease. What is the recommended treatment approach?

- **A:** Perioperative FLOT chemotherapy (4 cycles pre-op, surgery with D2 lymphadenectomy, 4 cycles post-op). Consider adding durvalumab if PD-L1 CPS 1.

Q2: Which molecular subtype of gastric cancer is most likely to respond to immune checkpoint inhibitors?

- **A:** MSI-H and EBV-positive subtypes, due to high mutational burden and prominent immune cell infiltration.

Q3: A patient with metastatic gastric cancer is found to be HER2-positive with PD-L1 CPS of 5. What is the preferred first-line regimen?

- **A:** Fluoropyrimidine + platinum + trastuzumab + pembrolizumab.

Q4: What is the most important modifiable risk factor for noncardia gastric cancer, and what is the evidence for intervention?

- **A:** *H. pylori* infection. Eradication therapy reduces gastric cancer incidence by ~46% in meta-analyses of RCTs.

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